Artificial intelligence in SOC alert triage: the moderating role of cyber hygiene, Zero Trust and tool fragmentation

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Each line is an alert; only a few are signal.

Abstract

The effect of artificial intelligence (AI) on alert triage in a Security Operations Center (SOC) depends on the quality of the environment in which it is deployed. This article proposes a conceptual framework for examining this thesis. Its research question is as follows: whether and how the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of AI, relative to conventional triage, on detection quality and on investigation cost.

The framework distinguishes information volume from information noise. It treats the use of AI as the focal variable, the three factors as moderators, and alert volume and rule configuration complexity as control variables. Four research propositions are formulated, each accompanied by its expected direction and its refutation criterion. A three-phase validation protocol is proposed: a reproducible targeted review, an exploratory qualitative study, and then a multilevel analysis complemented by a replay of labelled alerts.

The article reports no empirical results. It aims to provide a testable basis for determining under what conditions AI genuinely reduces information noise rather than automating existing deficiencies.

Keywords: SOC; alert triage; information noise; artificial intelligence; cyber hygiene; Zero Trust; tool fragmentation; moderation effect; human decision-making.

1. Introduction

SOCs concentrate the monitoring, detection, analysis and response functions for security incidents. They rely on solutions for security event management, endpoint detection, threat intelligence and response orchestration. The multiplication of cloud environments, the generalisation of digital identities and the growth of telemetry sources increase the volume of data to be processed, and organisations stack specialised tools to cover ever wider perimeters.

The literature describes the human consequences of this situation. Tariq et al. (2025) survey research on alert fatigue in SOCs. Alahmadi et al. (2022) and Kokulu et al. (2019) report, on the basis of interviews with analysts, the difficulties related to the relevance of alarms and to the fit between tools, processes and people. Vielberth et al. (2020) offer a systematic study of SOCs and their open challenges. Ofte and Katsikas (2023) examine operators' situational awareness.

Faced with these difficulties, AI is increasingly presented as a lever for prioritising, correlating and summarising alerts. Chhetri et al. (2024) study the cooperation between analysts and AI in this respect. However, AI is inserted into an environment that is already structured, with its detection rules, its tools and its security practices, and nothing allows us to assume that its contribution is identical from one SOC to another.

1.1 The gap addressed

The works cited most often examine alert load, human-AI cooperation or situational awareness. The relationship between the effect of AI and the quality of the technical and organisational environment is less formalised. The working hypothesis of this article is that this gap exists and that it is significant; the targeted review described in Section 5 must confirm or refute this before any submission.

A triage AI consumes context: asset identifiers, owners, account history, patch status, endpoint posture, logs from several tools. If this context is incomplete, inconsistent or scattered, the AI may go faster without deciding better. Conversely, a well-controlled environment may allow the AI to produce a real gain. Context is therefore not a mere confounding variable: it may condition the very effect one seeks to measure.

1.2 Research question

How do the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of artificial intelligence, relative to conventional triage, on detection quality and on investigation cost during alert triage in SOCs?

Three sub-questions break it down:

1. **SQ1 (hygiene).** Is the advantage of AI over conventional triage more pronounced when cyber hygiene is more mature?
2. **SQ2 (fragmentation).** Does this advantage diminish when tools are fragmented and poorly interoperable, or does AI on the contrary compensate for this fragmentation?
3. **SQ3 (Zero Trust).** For alerts related to identities and access, does a consistent Zero Trust implementation increase the advantage of AI?

Both outcomes are examined together: a time saving obtained at the price of missed threats does not constitute an improvement.

1.3 Contributions and outline

The article provides three elements: a functional definition of information noise distinct from volume and from alert fatigue; a moderation framework in which AI is the focal variable and three characteristics of the environment are moderators; and four falsifiable propositions with a validation protocol. Section 2 reviews the literature, Section 3 presents the framework, Section 4 formulates the propositions, Section 5 describes the validation method, and Sections 6 to 8 cover the discussion, the limitations and the conclusion.

2. Literature review

This review is deliberately focused on what underpins the model. It remains preliminary: Section 5 provides for a reproducible review before submission.

2.1 Alert volume and fatigue

Alert fatigue refers to the degradation of attention or vigilance that may appear when analysts handle numerous alerts whose relevance is low or difficult to establish (Tariq et al., 2025). The number of alerts measures load, not the difficulty of the work: two SOCs receiving the same number of alerts may face very different efforts depending on the available context, redundancy and investigation procedures. It is therefore necessary to distinguish the volume of events collected, the volume of alerts generated, the proportion of relevant alerts, the difficulty of qualification and the quality of final decisions.

2.2 AI in triage and human-AI cooperation

AI can prioritise alerts, link events, summarise context and automate repetitive tasks. Chhetri et al. (2024) argue for the value of a cooperation in which automation handles suitable tasks and human expertise remains engaged for ambiguous cases. Work on automation shows, however, that overreliance and complacency are known risks (Parasuraman and Riley, 1997; Parasuraman and Manzey, 2010). An AI that processes more alerts in less time therefore improves security only if the risk of missing a relevant threat does not increase. This caveat justifies measuring investigation cost and detection quality together.

2.3 Tool fragmentation and loss of context

SOCs combine several specialised technologies. This diversity improves coverage but may degrade integration: different identifiers for the same asset, redundant alerts, context to be reconstructed manually across several interfaces. Kokulu et al. (2019) document mismatches between technologies, processes and people in SOCs. The relevant variable is not the number of products, but interoperability, the semantic consistency of data and the sharing of context between components. An environment composed of many well-integrated tools is not fragmented in the sense of this model.

2.4 Cyber hygiene

Cyber hygiene encompasses the basic practices that keep an environment under control: asset inventory, patch management, administration of accounts and privileges, secure configuration. ANSSI (2017) offers a reference set of measures. For a SOC, these practices determine how easily an alert can be linked to an asset and to an owner, and the stability of baseline behaviours. An improvement in hygiene may nevertheless reveal previously invisible events and temporarily increase detections; the expected effect therefore concerns the interpretability of alerts, not their number.

2.5 Zero Trust

Zero Trust rests on explicit verification of access, least privilege and the evaluation of context and risk at each request (Rose et al., 2020). The CISA maturity model (2023) structures its implementation into pillars, including identity and devices. The associated controls produce information on identities, devices and access conditions which, when correlated with other sources, can help qualify an access alert. But poorly harmonised policies may also multiply denials and context changes. Zero Trust is therefore neither an intrinsic source of noise nor a guarantee of noise reduction. Its role in this model is restricted to alerts related to identities and access, where it provides context.

2.6 Summary and current support

The following table indicates, for each element of the model, the assumed role and the degree of documentary support currently available to this article. It highlights the parts to be consolidated by the targeted review.

Element	Assumed role in the model	Current documentary support
Alert fatigue and volume	Control variable	Strong (Tariq et al., 2025; Alahmadi et al., 2022)
Human-AI cooperation	Framework for assisted triage	Partial (Chhetri et al., 2024; Parasuraman and Manzey, 2010)
Tool fragmentation	Moderator	Partial, mostly qualitative (Kokulu et al., 2019)
Cyber hygiene	Moderator	Reference frameworks (ANSSI, 2017), little work on the link with the SOC
Zero Trust	Moderator, identity and access alerts	Reference frameworks (Rose et al., 2020; CISA, 2023), little work on the link with the SOC
Interaction between these factors and AI	Core of the contribution	To be established by the targeted review

The gap is therefore less the absence of literature on each of the factors than the absence, to our knowledge, of a framework that articulates them as moderators of the effect of AI on triage.

3. Conceptual framework

3.1 Definition of information noise

Information noise refers to the share of security information which, in a given operational context, mobilises analytical resources without contributing proportionately to the identification, qualification or management of relevant threats.

This definition is functional. A non-malicious alert is not necessarily noise: it may verify the coverage of a control or reveal a misconfiguration. Noise is judged in relation to the purpose of the investigation, the value of the signal and the effort required to exploit it. It is distinct from volume, which is a quantity, and from alert fatigue, which is a possible human consequence of repeated exposure to a load that is difficult to manage.

3.2 Model variables

The model poses a question of difference: does AI perform better or more cheaply than conventional triage, and does this difference vary with the environment? It retains one focal variable, three moderators, two outcomes and a set of control variables. The figure below gives the structure and the table the working definitions.

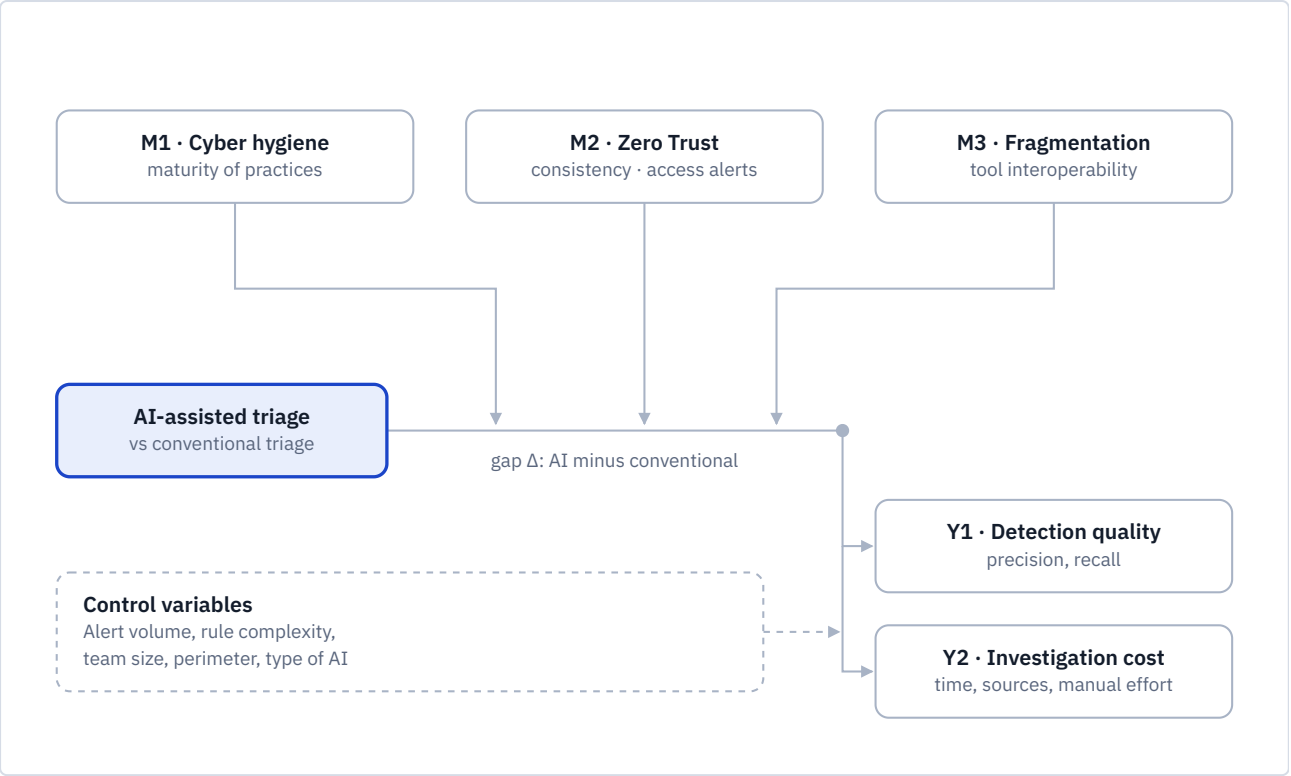


Figure 1. Moderation model: one focal variable, three moderators, two outcomes.

The three moderators act on the difference between AI and conventional triage, not directly on the outcomes; the control variables act on both conditions being compared.

Role	Variable	Working definition
Focal	AI-assisted triage	Use of AI to prioritise, correlate or summarise alerts, compared with conventional triage based on rules and manual procedures. Described by its functions, its level of automation and human supervision.
Moderator M1	Cyber hygiene maturity	Completeness of the asset inventory, patch compliance, control of accounts and privileges, maintained configurations.
Moderator M2	Consistency of Zero Trust implementation	Harmonisation of access policies and integration of identity and endpoint data. Applied to alerts related to identities and access.
Moderator M3	Tool fragmentation	Low effective interoperability: non-shared identifiers for the same asset, scattered context, number of interfaces to consult per alert.
Outcome Y1	Detection quality	Precision, recall and false-negative rate on relevant alerts.
Outcome Y2	Investigation cost	Qualification time, number of sources consulted, manual effort per alert.
Control	Volume and configuration	Alert volume, rule configuration complexity, team size and experience, monitored perimeter, type of AI.

3.3 Reading the model

Let Δ be the performance difference between AI-assisted triage and conventional triage, measured separately on Y1 and on Y2. A moderator is a variable that makes Δ vary: the effect of AI is not assumed to be constant, it is assumed to be conditional. The model does not say that AI is effective or ineffective in general; it says where and under what conditions its effect changes.

Three clarifications frame this reading.

1. **Maturity and consistency are two distinct constructs.** Hygiene is measured by a maturity of practices, Zero Trust by the consistency of an architecture and its policies. They will be operationalised separately (Section 5).
2. **Volume and rule complexity are controls, not objects of study.** They influence alert load in both conditions being compared, but the framework does not seek to explain their own effect.
3. **Rival readings exist.** Hygiene may also act as a factor that modifies the number of alerts (direct effect), and the most mature SOCs may be those that adopt AI first (selection bias). The protocol in Section 5 provides for within-SOC comparisons to limit this second risk.

4. Research propositions

The following four propositions are theoretical statements to be tested, not results. Each specifies the expected effect, the mechanism and what would refute it. The "AI advantage" refers, for each outcome, to the difference favourable to AI relative to conventional triage: higher detection quality or lower investigation cost.

P1 Baseline effect: the gain concerns cost first

Statement. On average, AI-assisted triage reduces investigation cost (Y2) relative to conventional triage, but this reduction is not sufficient to establish an improvement in detection quality (Y1), whose difference may be zero, positive or negative.

Mechanism. AI acts first on the speed of qualification through prioritisation and summarisation. Detection quality depends on the accuracy of rankings and on the risk of masking an important alert, a risk known in decision-support systems (Parasuraman and Manzey, 2010).

Refutation. P1 is refuted if cost is not lower under AI, or if the advantage on Y1 is uniformly positive across all environments, which would make it pointless to distinguish the two outcomes.

P2 Cyber hygiene: the advantage grows with maturity

Statement. With alert volume controlled, the AI advantage on Y1 and on Y2 is greater the higher the maturity of cyber hygiene (M1).

Mechanism. A complete inventory, controlled accounts and stable configurations allow AI to link each alert to an asset and to an owner, and to rely on stable baseline behaviours.

Caveat. Better hygiene may reveal previously invisible events and temporarily increase the number of detections; hence the control for volume.

Refutation. The interaction between AI use and M1 is zero or of opposite sign after controlling for volume.

P3 Tool fragmentation: the advantage decreases, or AI compensates

Statement. The AI advantage on Y1 and on Y2 is smaller the more tools are fragmented and poorly interoperable (M3).

Mechanism. Inconsistent identifiers and scattered context limit the ability of AI to link data; its outputs then rest on partial context.

Rival proposition. AI might on the contrary compensate for fragmentation by correlating scattered sources that the analyst would otherwise have to consult one by one. The advantage would then be more pronounced in a fragmented environment. These two readings are opposed and can be discriminated by the data; deciding between them is a possible contribution of the study.

Refutation. The interaction between AI use and M3 is zero or positive. A positive interaction would support the rival reading.

P4 Zero Trust: the advantage grows with consistency, for access alerts

Statement. For alerts related to identities and access, the AI advantage on Y1 and on Y2 is greater the more consistent the Zero Trust implementation (M2), that is, the more policies are harmonised and identity and endpoint data are integrated.

Mechanism. Zero Trust controls produce usable context for qualifying an access alert: endpoint posture, session risk, access decision applied.

Caveat. Poorly harmonised policies may multiply denials and context changes, and reduce the contribution of Zero Trust to triage.

Refutation. No interaction for these alerts, or an interaction of the same magnitude for alerts unrelated to access. This second case would indicate that the construct measured reflects general maturity rather than Zero Trust.

Summary

Proposition	Factor	Expected effect on the AI advantage	Alerts concerned
P1	None (baseline effect)	Advantage on Y2; advantage on Y1 not established	All
P2	M1 Cyber hygiene	Grows with maturity	All
P3	M3 Tool fragmentation	Decreases with fragmentation (rival reading: grows)	All
P4	M2 Zero Trust	Grows with consistency	Identities and access

The combination of the three moderators (higher-order interactions) is not formulated as a proposition: it is treated as an exploratory question in Section 7.

5. Proposed validation methodology

Validation follows a sequence of three phases, from the most exploratory to the most constraining. The same logic runs through it: comparing AI and conventional triage within contexts that differ in their hygiene, their Zero Trust and their fragmentation.

5.1 Phase 0. Reproducible targeted review

A restricted systematic review, reported according to PRISMA 2020 (Page et al., 2021), will verify the gap announced in Section 1.1. It will query the intersections between SOC, alert triage, AI, cyber hygiene, Zero Trust and tool integration, and will document the databases, the search strings, the inclusion criteria and the selection. Its result conditions the final formulation of the contribution.

5.2 Phase 1. Exploratory qualitative study

Semi-structured interviews will be conducted with SOC analysts, detection leads, CISOs and tool integration specialists. A target of around 12 to 20 interviews in 4 to 6 different SOCs (in-house, service providers, public and private sector) is proposed, until thematic saturation. The interviews will verify that the moderators correspond to real mechanisms and that the working definitions are operational, and will collect examples of cases where AI helped or hindered triage.

5.3 Phase 2. Multilevel analysis and controlled replay

Multilevel analysis. Alerts (level 1) are nested within SOCs or client perimeters (level 2). The moderators M1, M2 and M3 are measured at level 2; AI use varies within a single SOC where possible (before and after deployment, processing queues with and without AI). Mixed-effects models are estimated with cross-level interaction terms between level 1 and level 2 (Raudenbush and Bryk, 2002; Hayes, 2018). The power of level-2 interactions depends on the number of SOCs or perimeters, not on the number of alerts. If this sample size is insufficient to estimate three interactions, the study will retain two moderators, and managed service providers, which supervise several clients at different maturity levels, constitute a source of natural variation to be explored.

Controlled replay. A set of alerts labelled by experts serves as ground truth. Analysts process it with or without AI assistance, and the quality of the context provided is varied (complete or degraded inventory, shared identifiers or not, integrated identity data or not). This manipulation makes it possible to approximate a causal effect of the moderators on the AI advantage, which production data cannot establish on their own. Injected scenarios (for example derived from attack simulation exercises) complete the design to estimate recall, which is not observable in production.

5.4 Operationalisation

Construct	Indicators envisaged	Source
AI-assisted triage	Assisted functions, level of automation, human supervision, share of recommendations corrected	Interviews, tool logs
M1 Hygiene	Composite score inspired by ANSSI (2017) measures: inventory coverage, patch compliance, review of accounts and privileges	Audit, asset management data
M2 Zero Trust	Scale inspired by the identity and device pillars of the CISA (2023) model and the principles of Rose et al. (2020): policy consistency, data integration	Audit, architecture documentation
M3 Fragmentation	Number of interfaces consulted per alert, share of assets resolved under a common identifier, time to obtain context	Observation, logs
Y1 Detection quality	Precision, recall, false-negative rate	Labelled set, injected scenarios
Y2 Investigation cost	Median qualification time, number of cross-consultations, manual effort	Logs, timing
Controls	Alert volume, rule complexity, team size and experience, perimeter, type of AI	Operational data

The indicators will be fixed before data collection. A decrease in the number of alerts or in processing time will be interpreted as a gain only if recall is not degraded.

5.5 Analysis strategy

A descriptive analysis will characterise volumes, distributions and differences between contexts. The mixed-effects models will test the interaction between AI use and each moderator, with random effects by SOC and control variables; P4 will be estimated on identity and access alerts only, then compared with other alerts as a specificity test. Sensitivity analyses will vary the definition of the indicators, and multiple comparisons will be corrected. Since AI deployment is not randomised and the most mature SOCs may adopt it first, maturity is modelled explicitly and within-SOC comparisons are favoured. No result can be anticipated before data collection.

5.6 Ethics and confidentiality

Alert data are sensitive and may contain personal data. The study will require confidentiality agreements with the organisations, anonymisation or pseudonymisation of logs, and aggregated data for processing times so as not to evaluate analysts individually. Interviews will be subject to informed consent.

6. Discussion

6.1 Theoretical contributions

The framework shifts the research question from volume to the conditional value of AI. Instead of asking whether AI reduces alert fatigue, it asks in what context it improves detection and reduces investigation cost. This reformulation has three consequences.

1. **AI is treated as a processing capability embedded in an environment, not as an isolated solution.** Its effect is conditional, and it is this conditionality that is tested.
2. **A productivity gain is not a security gain.** By measuring Y1 and Y2 together, the framework prevents concluding that an AI performs well when it reduces processing time while letting more threats through.
3. **A rival reading is formulated from the outset.** For fragmentation, P3 and its rival proposition predict opposite signs, which makes the framework falsifiable on a point where practitioners' intuitions diverge.

6.2 Implications for cyber governance

Security and SOC managers can draw from the framework a diagnostic grid before investing in an AI layer. Do the observed difficulties stem from poorly calibrated rules, an incomplete inventory, poorly controlled accounts, non-interoperable tools or a lack of procedures? If the answer lies in the environment, AI risks producing a speed gain without a detection gain, and making it harder to spot the deficiencies it automates.

The framework also suggests supplementing volume indicators with measures of relevance, context, qualification cost and decision performance, and reserving ambiguous cases for human supervision. It does not oppose innovation: it seeks to determine the conditions under which innovation produces a measurable improvement.

6.3 Sequencing of investments

If P2 and P4 are borne out, cyber hygiene and Zero Trust consistency are not merely stand-alone security objectives: they increase the return on investments in detection AI. If P3 is supported, tool integration precedes AI. If it is the rival proposition that prevails, AI may on the contrary be a short-term palliative for fragmentation. In both cases, the investment decision benefits from relying on a diagnosis of the context rather than on the tool's promise alone.

7. Limitations and future directions

7.1 Limitations

- **Status of the manuscript.** The article is conceptual. It rests on no original data and its propositions are not established. The literature review is preliminary and must be replaced by the reproducible review of Phase 0.
- **Construct validity.** Hygiene, Zero Trust consistency and fragmentation are composite constructs whose measurement depends on context. The proposed scales are inspired by reference frameworks (ANSSI, 2017; CISA, 2023) that were not designed to measure moderators of AI effectiveness and will need to be validated.
- **Scope of Zero Trust.** The moderator M2 applies only to identity and access alerts. It says nothing about the triage of other families of alerts.
- **Ground truth.** Precision and recall require a reliable reference on relevant events. In production, recall is not observable and relies on injected scenarios, whose representativeness is limited.
- **Selection and causality.** AI deployment is not randomised. Within-SOC comparisons and the controlled replay limit the bias without eliminating it.
- **Heterogeneity of AI.** Machine-learning classifiers, correlation engines and assistants based on language models have neither the same strengths nor the same weaknesses, and evolve quickly. The type of AI is a control, but its results might not transfer from one generation of tools to the next.
- **Generalisation.** In-house SOCs, service providers and public bodies differ in their resources and objectives. Comparisons must take this into account.

7.2 Future directions

- **Higher-order interactions.** Do the effects of the three moderators accumulate, offset one another, or is one of them sufficient? The question is exploratory and requires a larger sample than that of a first study.
- **Temporal dimension.** Maturity evolves. A longitudinal study could track the AI advantage as hygiene, Zero Trust or tool integration improve.
- **Human factors.** The effect of human supervision, overreliance and complacency on Y1 could be modelled as an additional mediator or moderator.
- **Adversary adaptation.** A triage AI may become a target. Extending the framework to robustness against attacks aimed at detection deserves dedicated work.

8. Conclusion

This article proposes to shift the analysis of alert fatigue from volume to the conditional value of artificial intelligence. It formulates a precise question: whether and how cyber hygiene, Zero Trust consistency and tool fragmentation modify the effect of AI on triage, in terms of detection quality and

investigation cost.

The central proposition is that the operational value of AI in a SOC depends not only on its own capabilities, but also on the quality of configuration, integration and contextualisation of the environment in which it operates. Four falsifiable propositions, one of which pits two readings of the effect of fragmentation against each other, and a three-phase protocol make it possible to put it to the test.

If it is borne out, the question to ask before adding an AI layer is no longer only what the tool can do, but what the environment allows it to do. If it is refuted, the framework will at least have specified the conditions under which AI provides a measurable gain independently of context, which would also be a useful result.

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